AI 101 for Mechanical Engineers · Module 5

Investigate before changing readings

Lesson 5.1

What you will learn

> Investigate unusual and missing readings before changing them, explain what imputation does, and document changes without removing useful fault evidence.

Lesson overview

Module 5 · Lesson 1

Investigate before changing readings

AI 101 for Mechanical Engineers

1 Check the measurement context

Inspect values, timestamps, missing readings and physical plausibility.

Warning signs of a data problem

1. Negative absolute pressure
2. Duplicate timestamps
3. Mixed W and kW values

An outlier is unusual, not automatically wrong.
A high temperature may be a real fault event. Deleting it could remove useful evidence.

2 Missing values need investigation

A sensor may drop out during vibration or communication overload.

Missing reading → Investigate the cause → Choose and record a response

Imputation fills a gap using a stated rule.
It does not recover the original measurement.

Other rules: median, mode for categories, or interpolation for short gaps only when the process justifies it.

Record which values changed, the rule used and the reason.

3 Mean-imputation example

Original data

20	22	missing	24

Mean of the available values

(20 + 22 + 24) / 3 = 22

After replacement

20	22	22	24

An arithmetic illustration: no physical unit is stated.

Filling the gap makes the visible variation smaller.

Learn replacement values from training data only.
Apply the stored rule to validation and test data.

Key takeaway — **Investigate first. Keep useful fault evidence and document each change.**

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Screen measurements in context first

- Inspect distributions, timestamps, missing values and physical plausibility
- Warning signs: negative absolute pressure, duplicated timestamps, mixed W and kW
- Check what was measured and how it was recorded
- Only then select a correction

An outlier is not automatically an error

Outlier

- Unusual relative to a comparison population [27]
- A high temperature may be the fault event

Noise

- Variation or error that obscures the quantity of interest

Removing every high-temperature reading could remove important fault examples.

Missing readings can carry information

- A missing reading can carry information about the system [32]
- A sensor may drop out during vibration or communication overload
- Investigate the operating and measurement context
- Do not assume missingness is unrelated to the event

Imputation supplies a replacement, not the measurement

- Fills in or predicts a missing value using a stated method [32]
- It does not recover the original physical measurement
- Record which values changed, the rule used and why

Mean imputation fills the gap with 22

$$(20 + 22 + 24) \,/\, 3 = 22$$

- Readings [20, 22, missing, 24] become [20, 22, 22, 24]
- Fills the gap but artificially reduces visible variability
- No unit given: an arithmetic illustration only

Learn replacement values from training data only

- In a predictive pipeline, estimate replacements from training data
- Do not calculate them from the full dataset
- Do not refit them using final-test inputs

Match the response to the issue

Issue	Response
Missing reading	Investigate dropout; declare any replacement rule
High temperature	Keep operating context; investigate its significance
Mixed W and kW	Convert to consistent units before comparing
Imputed reading	Mark the value; record the rule

The replacement rule depends on why values are missing

Option	When it may suit
Median	Can be less affected by extreme values
Mode	A categorical field
Forward-fill or interpolation	Some short gaps, if measurement process justifies
Feature exclusion	Field mostly missing; compare with imputation

> Forward-filling and interpolation can mislead across a state change or long outage.

CHECK YOUR UNDERSTANDING

Why can removing all high-temperature readings damage a fault-detection dataset?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Why can removing all high-temperature readings damage a fault-detection dataset?

> Some readings may represent genuine fault events. Deleting them can remove the evidence the model is intended to recognize.

CHECK YOUR UNDERSTANDING

Does replacing the missing value with 22 recover the original measurement?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Does replacing the missing value with 22 recover the original measurement?

> No. It applies a mean-imputation rule; the original missing measurement remains unknown.

Key takeaways

- Investigate context before deleting unusual readings
- Missingness can itself be informative
- Imputation estimates replacements and can change visible variability
- Learn replacement parameters from training data; document every change

References

[27] V. Chandola, A. Banerjee, and V. Kumar, “Anomaly detection: A survey,” ACM Comput. Surv., vol. 41, no. 3, pp. 1–58, 2009. doi: 10.1145/1541880.1541882

[32] R. J. A. Little and D. B. Rubin, Statistical Analysis With Missing Data, 3rd ed. Hoboken, NJ, USA: Wiley, 2019. doi: 10.1002/9781119482260

Numbers match the course reference list. The written lesson is the authoritative text.